

COMP 3270 – Introduction to Algorithms

Homework 9

Topic – Graph Algorithms and BST

1. What is the difference between the binary-search-tree property and the min-heap property)? Can the min-heap property be used to print out the keys of an n-node tree in sorted order in $O(n)$? Show how or explain why not.
2. Suppose that we construct a binary search tree by repeatedly inserting distinct values into the tree. Argue that the number of nodes examined in searching for a value in the tree is one plus the number of nodes examined when the value was first inserted into the tree.
3. Show that if a node in a binary search tree has two children, then its successor has no left child and its predecessor has no right child.
4. Imagine you are designing a social network platform. How could you use Breadth-First Search to find the shortest connection path between two users on the platform? Provide a step-by-step explanation of how BFS would work in this context.
5. When searching for a file on a computer's file system, which graph searching algorithm (BFS or DFS) would you use to explore directories and locate the desired file? Provide an algorithmic explanation.
6. Consider the following matrix,

	A	B	C	D	E	F	G	H
A	0	1	0	0	0	1	0	0
B	0	0	1	0	1	0	0	1
C	0	0	0	1	0	0	0	0
D	0	1	0	0	0	0	0	1
E	0	0	0	1	0	0	1	0
F	0	0	0	0	1	0	1	0
G	0	0	0	1	0	1	0	0
H	0	0	0	0	0	0	1	0

- a. Draw the directed graph based on the matrix.
 - b. Perform a depth-first search on each of the following graphs. In cases where you have a choice of vertices to explore, prioritize the one that comes first in alphabetical order.
7. In the context of maze solving, with the goal of finding the shortest path from the start to the finish, which graph traversal algorithm would you choose: Breadth-First Search (BFS) or Depth-First Search (DFS)? Explain your choice based on the goals of minimizing path length, memory efficiency, and ease of implementation, providing a rationale for the selected algorithm.
 8. Given a graph with vertices and edges as,
 $V(G) = \{a, b, c, d, e, f, g\}$, $E(G) = \{ab, bc, bd, de, ef, fc, fg\}$
 - a) Find the BFS and DFS of the above graph and represent them as a tree.
 - b) Shows that if there is a cycle in graph G then the BFS tree and DFS tree are different.